

Cost Structures and Revenue Management

Chiu Yu KO

Low Altitude Economics

Week 3

Learning Objectives

By the end of this class, you should be able to:

- **Distinguish** CAPEX from OPEX in an LAE business model.
- **Calculate** simple unit-cost measures, including CASK and cost per mission-kilometre.
- **Estimate** breakeven volume and explain how price, load factor, and utilization affect viability.
- **Compare** pay-per-use, subscription, availability-based, and complementary revenue models.
- **Evaluate** how weather downtime and cost differences across Hong Kong and the GBA affect business strategy.

Midnight in Kowloon: The Numbers Behind the Service

Imagine two managers reviewing an LAE business case.

- **Manager A** asks how charging time, maintenance, and turnaround affect the number of daily missions.
- **Manager B** asks whether subscriptions or availability-based contracts could make revenue more predictable.

A technically feasible service is not necessarily commercially viable.

- Managers must connect asset cost, daily operations, capacity, demand, pricing, and risk.
- **Quick poll (2 min):** What is likely to be the most important cost driver for an LAE operator in Hong Kong?

Session 1

CAPEX vs OPEX, Unit Economics & Cost Drivers

Concept Primer: CAPEX and OPEX

CAPEX: Investment in assets that provide benefits over several years

- Examples: aircraft, charging equipment, vertiport construction, and major software systems.

OPEX: Recurring expenditure required to operate the service

- Examples: energy, maintenance, staff, insurance, site fees, communications, and routine compliance activities.

Important distinction: Certification and approval expenditure is not automatically CAPEX. Its accounting treatment depends on the nature of the expenditure and applicable accounting rules.

The “Hong Kong Premium”: Potential Cost Drivers

Why might an LAE service cost more to operate in Hong Kong?

- **Land and site adaptation:** Scarce sites, structural reinforcement, charging capacity, passenger access, and safety facilities may raise infrastructure cost.
- **Complex operating environment:** Dense buildings, terrain, communications, weather, and interactions with existing airspace users may increase operating requirements.
- **Labour and professional services:** Skilled staff, engineering, insurance, legal, and compliance support may be costly.
- **Reliability requirements:** Redundancy, maintenance capability, backup systems, and contingency planning can add cost.

These are cost hypotheses to test. The size of any Hong Kong premium will depend on the application, site, and operating model.

Applied Tool: Estimating Daily Economic Cost

A Simple Managerial Approximation

$$\text{Daily Economic Cost} = \frac{\text{Asset Cost} - \text{Expected Residual Value}}{\text{Useful Operating Days}} + \text{Daily OPEX}$$

- This spreads the net asset cost across its expected operating life.
- Daily OPEX includes both fixed operating costs and costs that vary with activity.
- Financing cost, maintenance cycles, asset downtime, and residual value may require separate treatment in a full model.

This is an economic-cost approximation, not a complete accounting depreciation or cash-flow model.

Back-of-the-Envelope: How Asset and Site Cost Affect Unit Cost

Assumptions: Five-year life, no residual value, daily OPEX of HK\$8,000, and 50 missions per operating day.

Case A: HK\$10 Million Investment

- Daily asset-cost allocation:
 $\text{HK\$10M} / 1,825 \approx \text{HK\$5,479}$
- Daily OPEX: HK\$8,000
- Total daily economic cost:
HK\$13,479
- **Cost per mission: about HK\$270**

Case B: HK\$6 Million Investment

- Daily asset-cost allocation: HK\$6M
 $/ 1,825 \approx \text{HK\$3,288}$
- Daily OPEX: HK\$8,000
- Total daily economic cost:
HK\$11,288
- **Cost per mission: about HK\$226**

Takeaway: Higher investment cost raises unit cost, but utilization and operating cost remain equally important.

Illustrative assumptions for classroom analysis; not current market quotations.

Concept Primer: “Unit Economics”

Unit economics examines the revenue and cost associated with one meaningful unit of activity.

- Passenger service: per flight, available seat-kilometre, or paying passenger-trip.
- Cargo or public-service operation: per mission, mission-kilometre, or kilogram delivered.
- **CASK**: Operating cost divided by available seat-kilometres. It measures the cost of capacity offered, whether seats are occupied or not.
- **Cost per mission-kilometre**: A simple mission-based measure for comparing cargo, inspection, and public-service operations.

Different metrics answer different managerial questions and should be interpreted together with demand and service quality.

Applied Tool: Calculating CASK

Cost per Available Seat-Kilometre

$$CASK = \frac{\text{Operating Cost for the Period}}{\text{Available Seats} \times \text{Kilometres Flown}}$$

- Available seat-kilometres measure the passenger capacity offered, not the number of seats sold.
- More productive aircraft use can spread some fixed operating costs over more seat-kilometres.
- A low CASK does not guarantee profitability. The operator must also fill enough seats and earn sufficient revenue.
- Turnaround, charging, maintenance, weather, route length, and available capacity all affect the measure.

Back-of-the-Envelope: How Utilization Affects CASK

Assumptions: Four available seats and total operating cost of HK\$12,000 per day.

10 Trips of 10 km

- Available seat-km:
 $4 \times 10 \times 10 = 400$
- $CASK = 12,000/400$
- **CASK = HK\$30 per seat-km**
- Allocated cost of one available seat on a 10 km trip: HK\$300

50 Trips of 10 km

- Available seat-km:
 $4 \times 50 \times 10 = 2,000$
- $CASK = 12,000/2,000$
- **CASK = HK\$6 per seat-km**
- Allocated cost of one available seat on a 10 km trip: HK\$60

Takeaway: Higher utilization lowers CASK in this simplified example, but does not establish market demand or profitability.

Illustrative assumptions for classroom analysis.

Applied Tool: Cost per Mission-Kilometre

A Mission-Based Cost Measure

$$\text{Cost per Mission-Km} = \frac{\text{Operating Cost for the Period}}{\text{Total Mission-Kilometres}}$$

- Useful for comparing cargo, inspection, emergency, or public-service operations.
- Lower cost can come from reducing operating cost or increasing productive mission volume.
- Longer missions mechanically create more mission-kilometres, so comparisons should consider route length, payload, service quality, and mission complexity.

Back-of-the-Envelope: Mission Volume and Unit Cost

Assumptions: Daily operating cost = HK\$12,000; average mission distance = 8 km.

20 Missions per Day

- Mission-km: $20 \times 8 = 160$
- Cost per mission-km: HK\$75
- **Allocated cost per mission: HK\$600**

100 Missions per Day

- Mission-km: $100 \times 8 = 800$
- Cost per mission-km: HK\$15
- **Allocated cost per mission: HK\$120**

Takeaway: Greater productive volume can reduce allocated unit cost, but capacity, demand, turnaround, and service requirements constrain volume.

Illustrative assumptions for classroom analysis.

Session 2

Revenue Streams, Yield Management & Breakeven

Moving Beyond a Single Transaction

LAE businesses may combine several revenue models:

1. **Pay per use:** The customer pays for each passenger trip, delivery, inspection, or mission.
2. **Subscription or membership:** The customer pays periodically for access, priority, discounts, or a defined service allowance.
3. **Availability-based contract:** A public agency or business pays for readiness and agreed service levels, with possible additional usage payments.
4. **Complementary services:** Data, analytics, maintenance, platform, or integration services may create additional revenue, subject to consent, data rights, privacy, and customer demand.

Choosing a Revenue Model

Model	Revenue predictability	Usage risk	Possible
Pay per use	Lower	Operator	Passenger
Subscription	Medium	Shared	Frequent
Availability contract	Higher	Customer / shared	Public or c
Complementary service	Varies	Varies	Data or pla

- Match the revenue model to the value promised and the party best able to manage the risk.
- A subscription is not guaranteed revenue unless customers accept the contract terms and continue paying.
- Data does not generate revenue automatically. It requires useful information, data rights, customer demand, security, and privacy compliance.
- Combining models may diversify revenue but also increases contractual and operational complexity.

Concept Primer: Revenue Management

Revenue management allocates limited capacity and adjusts prices or conditions to improve total expected revenue.

- The objective is not simply to eliminate every empty seat.
- Selling too cheaply may fill capacity while reducing total revenue.
- Holding capacity for later high-value demand can sometimes be worthwhile.
- Managers consider price, booking time, customer segment, expected demand, cancellation risk, and capacity.

Managerial objective: Sell the right capacity to the right customer at the right time and price.

Applied Tool: The Breakeven Equation

One Service Unit

$$\text{Breakeven Units} = \frac{\text{Fixed Cost}}{\text{Revenue per Unit} - \text{Variable Cost per Unit}}$$

- The denominator is the **contribution margin per unit**.
- The unit must be consistent: one flight, one passenger, one delivery, or one mission.
- For passenger operations, revenue per flight depends on both average fare and paying passengers per flight.

Back-of-the-Envelope: Price and Breakeven Volume

Assumptions: Fixed cost = HK\$20,000/day; variable cost = HK\$100 per mission.

Revenue of HK\$500 per Mission

- Contribution margin: HK\$400
- Breakeven missions: 20,000/400
- **50 missions per day**

Revenue of HK\$200 per Mission

- Contribution margin: HK\$100
- Breakeven missions: 20,000/100
- **200 missions per day**

Takeaway: Discounting can sharply increase required volume. The manager must test whether the target fits available capacity and expected demand.

Illustrative assumptions for classroom analysis.

Applied Tool: Required Daily Activity

A Breakeven Activity Target

$$\text{Required Missions per Day} = \frac{\text{Fixed Daily Cost}}{\text{Contribution Margin per Mission}}$$

- Compare the required activity with physical capacity, operating hours, charging, maintenance, weather, and expected demand.
- This is not the usual definition of a utilization rate. A utilization rate is normally actual productive use divided by available capacity or time.
- Revenue management can improve contribution, but cannot create unlimited capacity or demand.

A Simple Passenger-Revenue Relationship

$$\text{Revenue per Flight} = \text{Seats} \times \text{Load Factor} \times \text{Average Fare}$$

- Load factor is the proportion of available seats occupied by paying passengers.
- Average fare should reflect discounts, customer mix, and booking conditions.
- The objective is to improve total contribution, not merely to maximize load factor.

Back-of-the-Envelope: Load Factor and Breakeven

Assumptions: Four seats, average fare HK\$500, fixed cost HK\$20,000/day, and variable cost HK\$100/flight.

40% Load Factor

- Expected paying passengers:
 $4 \times 40\% = 1.6$
- Revenue per flight:
 $1.6 \times 500 = \text{HK\$}800$
- Contribution: HK\$700
- Breakeven: $20,000/700 \approx \mathbf{29}$ flights/day

90% Load Factor

- Expected paying passengers:
 $4 \times 90\% = 3.6$
- Revenue per flight:
 $3.6 \times 500 = \text{HK\$}1,800$
- Contribution: HK\$1,700
- Breakeven: $20,000/1,700 \approx \mathbf{12}$ flights/day

Takeaway: Higher load factor can reduce the required number of flights, provided fares and costs remain unchanged.

Illustrative assumptions; fractional passengers represent an average across many flights.

Session 3

Cost Sensitivities, Risk Hedging & GBA Scale

Concept Primer: Weather Downtime

Severe weather can affect an LAE business through several channels:

- lost trips and revenue;
- continued aircraft, site, staff, insurance, and financing costs;
- customer disruption, refunds, and service-level penalties;
- repositioning, inspection, maintenance, and recovery costs; and
- lower customer willingness to rely on the service.

A warning signal does not mechanically determine every aviation operation. Actual restrictions depend on aircraft capability, location, operational approval, and prevailing conditions.

Concept Primer: Managing Weather Risk

Managers can combine operational, contractual, and financial responses:

- **Operational resilience:** Forecasting, spare capacity, alternative routes or modes, and recovery planning.
- **Contract design:** Availability payments, minimum commitments, cancellation terms, service credits, and force-majeure provisions.
- **Insurance or parametric cover:** A payment may be triggered by an agreed event or weather measure, subject to availability, exclusions, basis risk, and premium cost.
- **Liquidity planning:** Cash reserves or committed financing to absorb temporary revenue shortfalls.

Subscriptions and reservation fees transfer risk only if customers accept the terms and receive a credible value proposition.

Applied Tool: Profit Under a Mixed-Revenue Model

A Simple Monthly Model

$$\text{Profit} = \text{Committed Revenue} + \text{Usage Revenue} + \text{Insurance Payout} \\ - \text{Fixed Cost} - \text{Variable Cost} - \text{Premiums and Service Credits}$$

- Committed revenue may reduce exposure to flight volume, but often comes with availability or service obligations.
- Downtime may save some variable cost while still creating refunds, credits, recovery costs, or reputational damage.
- The objective is not to profit from disruption. It is to keep cash flow and service obligations manageable.

Back-of-the-Envelope: A Month with Five Grounded Days

Assumptions: 30 days; 25 operating days; 50 missions/day; HK\$500 usage revenue; fixed cost HK\$600,000/month; variable cost HK\$100/mission.

Pay per Mission

- Missions: 1,250
- Revenue: HK\$625,000
- Variable cost: HK\$125,000
- Fixed cost: HK\$600,000
- **Operating loss: HK\$100,000**

Committed Monthly Revenue

- Contract revenue: HK\$750,000
- Variable cost: HK\$125,000
- Fixed cost: HK\$600,000
- **Operating profit before service credits: HK\$25,000**

Takeaway: Committed revenue may stabilize cash flow, but the contract must specify availability, refunds, service credits, and weather-related obligations.

Illustrative assumptions; excludes taxes, financing, insurance premiums, recovery costs, and possible service credits.

Hong Kong and the GBA: Different Cost and Revenue Possibilities

Dimension	Hong Kong possibility	Wider GBA possibility
Site environment	Dense and constrained	More varied
Customer base	Premium and specialized	Larger and more diverse
Cost pressure	Site and professional costs	Varies by city and use case
Scale opportunity	Complex urban services	Manufacturing and network scale
Potential role	Finance, logistics, services	Technology, production, operation

These are strategic hypotheses rather than universal facts. Smart operators should compare specific sites, routes, customers, regulations, and partners.

In-Class Exercise: Spreadsheet Workshop

Activity: Unit Economics & Sensitivity Testing

- **Format:** Groups of 4–5 (Laptops required)
- **Scenario:** You are managing a new HK Airport Shuttle fleet.
- **Task:** Using the provided Excel template:
 1. Input the Base Case assumptions to find your Breakeven.
 2. **Shock 1:** Increase Vertiport Rent by 20%.
 3. **Shock 2:** Simulate 3 days of Typhoon downtime.
 4. How much do you have to raise ticket prices to survive?
- **Timing:** 15 minutes hands-on + 5 minute share-out.

Exercise Template (Visual)

[Distribute digital Excel Spreadsheet to student groups]

The Dashboard View:

Input: Fleet Size | Daily Rent | Electricity Cost | Ticket Price

Output: Daily Profit | Breakeven Flights | CASK

Watch how the "Daily Profit" box turns red the second you lower the utilisation rate!

Key Takeaways

- Separate CAPEX, recurring fixed OPEX, and activity-related variable cost before calculating unit economics.
- CASK measures the cost of passenger capacity offered; mission-based measures may be more useful for cargo and public-service applications.
- Breakeven depends on a consistent unit, contribution margin, feasible capacity, and realistic demand.
- Revenue management aims to improve total contribution, not simply maximize load factor.
- Subscriptions, availability contracts, data services, insurance, and liquidity can reshape risk, but none provides a free or guaranteed hedge.
- Hong Kong and other GBA cities may offer complementary cost, customer, technology, and service advantages.

Week 3 Assignment Briefing

Cost and Revenue Model with Memo (maximum 1,200 words plus spreadsheet)

- **Due:** Beginning of Week 4 (group assignment)
- **Task:** Extend the Week 2 corridor or application into a simple operating model.
- **Requirements:**
 1. Separate investment cost, fixed OPEX, and variable cost.
 2. Calculate one appropriate unit-cost metric and breakeven volume.
 3. Model revenue per mission using price, capacity, and expected utilization or load factor.
 4. Test three shocks: lower price, weather downtime, and higher compliance or infrastructure cost.
 5. Recommend one operational, contractual, or financial risk response and discuss its limitation.
- **Grading focus:** Accurate and internally consistent model (40%), transparent assumptions (30%), actionable recommendation (20%), professionalism (10%).

Demand Modeling in Urban Air Mobility

Preview question for next week:

Now that we know exactly how much we HAVE to charge to survive... how do we mathematically prove that customers are actually willing to pay it?

Thank You + Q&A

Pre-Class Readings Reminder (for Week 4):

- *McKinsey: Perspectives on AAM (Adoption patterns)*
- *NASA UAM Market Study (Trip generation & Mode choice)*

Questions?

(Final 15 minutes reserved for extended Q&A)